Ameer Mashahreh

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PROFESSIONAL SUMMARY

Innovative Computer Engineering student with demonstrated expertise in building multi-language, AI-powered emergency communication systems. Lead developer of QIEM, a quantum-inspired MANET network for disaster response, integrating 6 programming languages (Python, Rust, C++, Go, JavaScript, Julia). Proven ability to design cybersecurity threat detection systems, real-time 3D visualizations, and intelligent routing algorithms. Seeking software engineering or cybersecurity roles where technical versatility and system integration skills can drive impactful solutions.

CORE COMPETENCIES

Programming Languages:

Python (Advanced), Rust (Intermediate), C++ (Intermediate), Go/Golang (Intermediate), JavaScript (Intermediate), Julia (Foundational)Frameworks & Tools: Streamlit, Three.js, PyDeck, Plotly, OpenCV, Git & GitHub, SQLite, WebSocketsCybersecurity & AI: Intrusion Detection, Isolation Forest, Random Forest, Q-Learning, Anomaly Detection, Threat Classification, Self-Healing Networks, MANET Routing

System Architecture: Distributed Systems, Real-Time Data Streaming, Multi-Language Integration, API Development, Database Design, 3D Visualization

Soft Skills: Technical Leadership, System Integration, Complex Problem-Solving, Team Collaboration, Project Management, Technical Documentation, Strategic Planning

EDUCATION

Bachelor of Science in Computer Engineering **Expected Graduation: 2030**
Al-Quds University | Abu Dis, Palestine

Relevant Coursework: Cybersecurity Fundamentals, AI Foundations, Data Structures, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks

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PROJECTS

QIEM – Quantum Intelligent Emergency Mesh Network

GitHub: github.com/ameermashahrh/qiem-network-c2

- Engineered a revolutionary disaster-response communication system that operates without Internet or cellular infrastructure, using self-organizing MANET (Mobile Ad-hoc Network) of drones and mobile nodes.
- Architected a multi-language codebase integrating 6 programming languages: Python (Streamlit UI, AI models), Rust (fast-path computation, 2-5ms latency), C++ (Dijkstra optimization), Go (WebSocket server for live data streaming, <100ms response), JavaScript (Three.js 3D interactive mapping), and Julia (predictive analytics).
- Designed and implemented quantum-inspired routing algorithms and AI-driven path optimization using Q-Learning, achieving 10x faster path calculation compared to pure Python implementations.
- Developed a real-time cybersecurity threat detection system capable of identifying and isolating 5 types of attacks (Blackhole, Sybil, Grayhole, Wormhole, Cascade Failure) using Isolation Forest and Random Forest algorithms with automated self-healing capabilities.
- Built an interactive command center dashboard featuring 2D/3D network visualization (PyDeck & Three.js), live telemetry monitoring, battery drain prediction using Linear Regression, and intelligent network optimization recommendations.
- Created a comprehensive academic reporting system generating detailed PDF, Excel, and CSV exports for documentation and operational analysis.- Implemented a multi-role access control system (Admin, Operator, Analyst, Viewer) with Arabic/English language support for diverse user scenarios.

NEXUS Smart Stadium Management SystemGitHub:

github.com/ameermashahrh/smart-stadium

- Developed a futuristic smart stadium management platform with real-time ticketing, access control, and venue operations management.
- Integrated QR code technology for ticket validation, reducing manual entry time by an estimated 60%
- Built interactive 3D digital twin visualization of stadium infrastructure for real-time monitoring and operational control.
- Implemented live Excel database synchronization for dynamic seat management and availability tracking.
- Designed role-based access controls and cyber-security monitoring features for stadium operations.

MedicalVault – Medical Records Security System

GitHub : github.com/ameermashahrh/MedicalVault

- Developed a Python-based secure application for managing electronic medical records with emphasis on patient data privacy.
- Implemented robust user authentication protocols and role-based access control to protect sensitive health information.
- Built a secure SQLite database backend with 100% test coverage for authentication modules.

CERTIFICATIONS & TRAINING

- Python Projects Development - ZyTek Engineers (OOP, File Handling, Git & GitHub)
- -Cybersecurity Bootcamp – Handeshah Sah Club (August 2025)
- - Cybersecurity & Bug Bounty Bootcamp – Society of Engineers and Developers (SED) (2025)
- - AI Foundations & MVP Bootcamp – BinX Tech (10-hour intensive program)
- Ethical Hacking & Kali Linux Course – ZyTek Engineers (August 2026) — Fundamental ethical hacking, penetration testing, and Kali Linux tools.
- - Excel and AI Usage Course – Community Knowledge & Research Library (24-hour training)

ACHIEVEMENTS

- - Lead developer and system architect of QIEM, a multi-award eligible disaster-response network prototype.
- - University Ambassador for Bin X Tech, representing the company at Al-Quds University events and building communication channels between students and the organization (2025-Present).
- - Successfully integrated 6 programming languages in a single production-ready system demonstrating cross-platform technical versatility.
- - Designed a self-healing network capable of automatically isolating compromised nodes within milliseconds of attack detection.

ADDITIONAL INFORMATION

- Active member of multiple tech communities and bootcamps, demonstrating commitment to continuous learning and professional development.
- Passionate about applying technology to solve humanitarian challenges, particularly in emergency response and disaster management.
- Available for internship or junior-level positions in Software Engineering, Cybersecurity, or AI/ML Development.